THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 2 OF 4

HANDS-ON MATH LAB

Building Mathematical Understanding with Physical Blocks, Paper-First Problem Solving & Stack Verification

A self-contained, open-and-go K–12 student exploration workbook for homeschooling families and classrooms. Connects tactile manipulative models—Place Value Studio (PLACE-01), Fraction Studio (FRAC-01), and Prime Factor Studio (PRIME-01)—with deep mathematical concepts, paper-and-pencil predictions, and distraction-free self-checking on the four-line StackCalc SC-6 and SC-32 calculators. Works alongside any homeschool or school math curriculum.

StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems) & 15-Page Coach Guide
- Book 4 — Stack Speed & Fluency: 15 Speed Lessons, Shortcut Playbooks & 60 Progressive Practice Tests (1,200 Problems)

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio (PLACE-01): 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio (FRAC-01): 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio (PRIME-01): Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

COPYRIGHT, STATE STANDARDS ALIGNMENT & OPEN-EDUCATION LICENSE

HANDS-ON MATH LAB: Building Mathematical Understanding with Physical Blocks, Paper-First Problem Solving & Stack Verification

Copyright © 2026 by StackCalc (https://stackcalc.io). Some rights reserved. Published by StackCalc Learning Lab, San Francisco, California. First Edition (US Letter 8.50 × 11.00 in / 215.9 × 279.4 mm).

State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

Open-Education License (CC BY-NC-SA 4.0) & Perpetual Classroom Reproduction Permission:

Classroom teachers, math coaches, interventionists, homeschool families, and schools are expressly permitted to photocopy or print student lab worksheets, speed trials, reference cards, and parent welcome letters for non-commercial educational instruction under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International.

Regulatory & Physical Safety Notice (Reese's Law / P.L. 117-171 / 16 CFR Part 1263 & ASTM F963):

Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

Trademark Notice & Print Colophon:

StackCalc, SC-32, and SC-6 are trademarks of StackCalc. All curriculum and textbook names are trademarks of their respective owners, none of whom endorse or sponsor this independent publication. Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, and STIX Math in high-contrast monochrome ink on archival white paper.

Table of Contents

Check off each unit as you complete the Guided Walkthrough and the 4-Problem Lab Investigation.

Section / Unit & Mathematical Topic	Grades & Calculator	Real-World Mathematical Context	Pages / Done
Preface: Welcome to the Hands-On Math Lab!	Students & Families	Paper & Blocks First, 3 Kits & 2-Min Warm-Up	**p. iv [ ]**
Unit 01: Cardinality, Counting & Step Accumulation Kit: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens)	Pre-K – Kindergarten (SC-6)	How Ten Fingers & an Empty Shell Created Place Value	**pp. 1–2 [ ]**
Unit 02: Part-Part-Whole Number Bonds & Missing Addends Kit: Place Value Studio (Two-Row Tray & Digit Blocks 0–9)	Grades K–1 (SC-6)	Part-Part-Whole Number Bonds & Friends of Ten	**pp. 3–4 [ ]**
Unit 03: Complements to 10 & Gold [+1] Carry Regrouping Kit: Place Value Studio (Whole-Number Tray, Digit Blocks 0–9 & Gold [+1] Carry Tokens)	Grade 1 (SC-6)	Making a Ten First (8 + 5 = 10 + 3 = 13) & The 10-for-1 Carry Trade	**pp. 5–6 [ ]**
Unit 04: Twin-Row Place Value Tray: Doubles & Equal Groups Kit: Place Value Studio (Two-Row Tray & Digit Blocks 0–9)	Grades 1–2 (SC-6)	Symmetry & Twin Rows: Why Doubles Anchor Mental Math	**pp. 7–8 [ ]**
Unit 05: Place Value & The 4-Level Stack Elevator Kit: Place Value Blocks & 4-Register Stack	Grades 3–4 (SC-6)	Inside-Out Multi-Step Problem Solving & Young Gauss's Pairing Insight	**pp. 9–10 [ ]**
Unit 06: Fraction Studio: 84 mm Towers & Two-Dot Entry Kit: 84 mm Fraction Towers & Twin Comparator Well	Grades 4–5 (SC-6)	The Pharaoh's Bakery & Why the Whole Fraction Tower Stands 84.0 mm Tall	**pp. 11–12 [ ]**
Unit 07: Postfix Trees, Order of Operations & Prime Factors Kit: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack	Grades 5–6 (SC-6)	Why 24 Is the Champion Packing Number & How Inside-Out Evaluation Eliminates Parentheses	**pp. 13–14 [ ]**
Unit 08: Base-x Polynomials, Prime Factor Pairs & Roots Kit: Prime Factor Studio (PRIME-01) & Algebraic Area Diagram	Grades 7–8 (SC-32)	Al-Khwarizmi's Geometric Squares & The 1535 Public Algebra Duel in Milan	**pp. 15–16 [ ]**
Unit 09: Coordinate Geometry, Trigonometry & Vectors Kit: Right-Triangle & 2D Vector Diagram (Optional Ratio Check: Fraction Studio)	Grades 9–11 (SC-32)	A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow	**pp. 17–18 [ ]**
Unit 10: Exponents, Logarithms & Continuous Growth Kit: Prime Factor Studio (PRIME-01 Logarithmic Blocks) & Growth Curve	Grades 10–12 (SC-32)	John Napier's Logarithms: Turning Multiplication into Height Addition	**pp. 19–20 [ ]**
Unit 11: Combinatorics, Probability Ratios & Series Statistics Kit: Prime Factor Studio (PRIME-01 Factorial Cancellation) & Statistical Data Plot	Grades 11–12 / AP (SC-32)	Sir Francis Galton's 1873 Bean Machine & The Mathematics of Small Samples	**pp. 21–22 [ ]**
Unit 12: Digital Logic, Base-N & Computer Architecture Kit: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns)	Grades 9–12 / AP CS (SC-32)	Leibniz's Binary Medallion & Claude Shannon's Boolean Relay Circuits	**pp. 23–24 [ ]**

APPENDICES A & B: STUDENT SELF-CHECK ANSWER KEY (PAGE 25) & MASTERY CERTIFICATE (PAGE 26)

Complete all 12 units and check your work on the SC-6 or SC-32 calculator to earn your Stack Mastery Certificate!

Preface: Welcome to the Hands-On Math Lab!

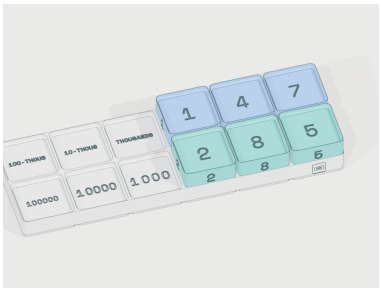
Why we build with blocks and solve on paper first—and how your distraction-free stack calculator checks your work.

1. PAPER & BLOCKS FIRST: WHY THIS LAB MAKES YOU STRONGER AT MATH (NEVER A CRUTCH!)

Whether you are learning at your homeschool kitchen table or in a classroom, this workbook was built on one simple rule: a calculator should NEVER replace your brain! In every unit, you first touch and build the problem with physical blocks (or sketch a visual model) and solve it by hand on paper. Only after you write your paper prediction do you pick up your distraction-free StackCalc calculator to check your work.
Why does your calculator have an [ENTER] key instead of [=]? Think about how you add 8 + 5 on paper: first you write 8 on the top line (Waiting Shelf, Line Y). Next you write 5 below it (Workbench, Line X). Only after both numbers are in place do you add (+) to get 13! On your 4-line Stack screen (T, Z, Y, X), you press [8] [ENTER] to put 8 on Shelf Y, type [5] on Workbench X, and press [+] to combine them—just like paper math!

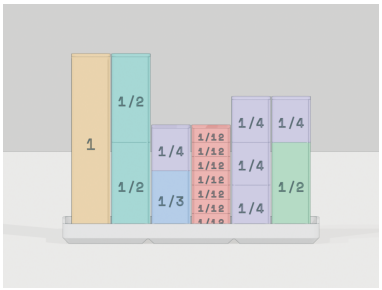
2. YOUR THREE HANDS-ON BLOCK KITS (SCREEN-FREE TACTILE DISCOVERY)

1. Place Value Blocks (PLACE-01)



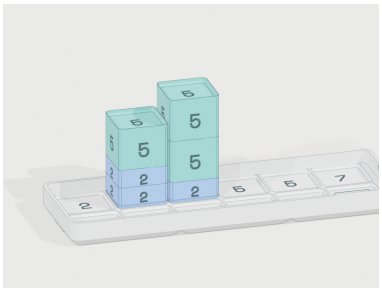
2-Row Tray (Y & X), Digits 0–9 & [+1] Carry

2. Fraction Towers (FRAC-01)



84 mm Proportional Towers (1 to 1/12)

3. Prime Factor Blocks (PRIME-01)



Logarithmic Prime Blocks (2, 3, 5, 7)

3. OPEN-AND-GO FOR HOMESCHOOLS & CLASSROOMS (2-MINUTE WARM-UP)

• 100% Self-Contained & Curriculum-Friendly: You do not need any other book to complete these 12 labs! They pair naturally with Singapore Math, Saxon, RightStart, Math-U-See, Beast Academy/AoPS, or any school textbook. You never write out button lists—you write real math steps on paper and verify on the stack.
• Inside-Out Stack Check: Floor 1 (Line X) is your active Workbench; Floor 2 (Line Y) is your Waiting Shelf; Floors 3 & 4 (Z & T) hold earlier steps so you can verify multi-step problems without parentheses!

• **Try Addition on Your Stack (8 + 5 = 13):** [8] [ENTER] [5] [+]

• **Try a Two-Dot Fraction (1/2 + 1/4 = 3/4):** [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

The 4-Step Lab Loop in Every Unit: 1. Mathematical Concept → 2. Blocks / Model → 3. Paper Prediction → 4. Check!

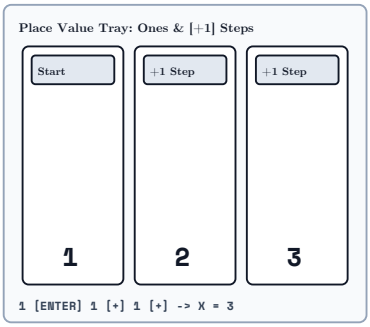
Always solve and write your prediction on paper before pressing calculator keys! After finishing the 4 lab investigations on the right-hand page of any unit, check your answers in Appendix A (Page 25).

Counting, Cardinality & The One-More Step Counter

Hands-On Model: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens) • Standards: CCSS.MATH.K.CC.A.1, CCSS.MATH.K.CC.B.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (1-to-1 Correspondence: Stepping Up by One):

Counting by ones is repeated addition of the unit 1 (n → n + 1). On the Place Value Tray and the stack, your current count waits first while you bring in 1 more step, then both combine into the new total.



2. HANDS-ON BLOCKS & CONCEPT RULES: HOW COUNTING & PLACE VALUE WORK TOGETHER

- Numbers show quantity (cardinality): the Digit Block in the Ones socket shows how many items you have counted.
- Adding 1 more item steps the count up by 1 (n → n + 1): 1 becomes 2, 2 becomes 3, and 3 becomes 4.
- An empty tray has zero items, represented by the [0] Digit Block in the Ones socket.
- When checking with a partner or teacher on the SC-6, Line Y holds your starting count and Line X adds the next [+1] step.

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place the [1] Digit Block into the Ones socket of the Place Value Tray. Keep Gold [+1] step tokens ready beside the tray. Question: When we place 1 Gold [+1] step token and step the Ones block up by 1, what digit block replaces [1]? (Prediction: [2])

Verification Keystrokes: [1] [ENTER] [1] [+] [1] [+]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[1]	1	First count placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Gold [+1] step token added: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Swap Ones block to [2]
[1]	1	Next Gold [+1] step token added: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Tray [3] and screen match perfectly

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Place the [4] Digit Block in the Ones column of your Place Value Tray and add 1 Gold [+1] step token. What total quantity do you have now?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Row 1 holds [6] and Row 0 holds [2] on your Place Value Tray. Which row has more, and how many more does it have?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 01 Lab: Counting, Cardinality & The One-More Step Counter

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: Place [3] in the top row and [2] in the bottom row of your Place Value Tray. Combine both groups—what is the total?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: You have 5 ones on the tray and take away 2. What Digit Block shows how many ones remain?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: You have 3 blocks on the tray and take away all 3. How many blocks are left, and which Digit Block shows an empty tray?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Start with [1] in the Ones socket and add one more [+1] token three times in a row (1 + 1 + 1 + 1). What count do you reach?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

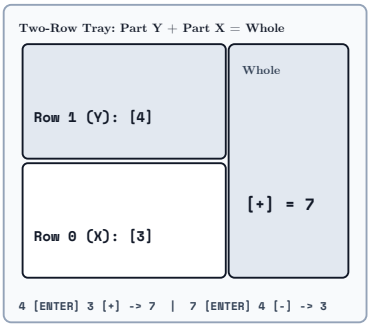
EXIT REFLECTION: When you add one more [+1] token to a full tray of 9, how do the Tens and Ones sockets change?

Part-Part-Whole & Number Bonds on the Place Value Tray

Hands-On Model: Place Value Studio (Two-Row Tray & Digit Blocks 0–9) • Standards: CCSS.MATH.K.OA.A.3, CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Composing Numbers: Part + Part = Whole):

Every addition problem composes two independent parts into a single whole (Part 1 + Part 2 = Whole). Placing Part 1 on the top row (Shelf Y) and Part 2 on the bottom row (Shelf X) makes both parts visible before they are joined.



2. HANDS-ON BLOCKS & CONCEPT RULES: PART-PART-WHOLE NUMBER BONDS ON THE TWO-ROW TRAY

- Row 1 (top row) of the Place Value Tray holds Part 1, and Row 0 (bottom row) holds Part 2.
- Putting both parts together (Part 1 + Part 2) composes the Whole.
- To find a missing part when you know the Whole: start with the Whole and subtract the known Part (Whole - Part 1 = Part 2)!

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place the [4] Digit Block in Row 1 (Part 1) and the [3] Digit Block in Row 0 (Part 2) of the Ones column on your Place Value Tray. Question: Combine both parts in the Ones column: 4 + 3 = 7 (swap to the [7] Digit Block).

Verification Keystrokes: `4` `ENTER` `3` `+`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`4`	**4**	Part 1 (4) enters register X
`ENTER`	**4**	4 copied up to register Y; represents Row 1 (Part 1)
`3`	**3**	Part 2 (3) enters register X; represents Row 0 (Part 2)
`+`	**7**	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Place [5] in Row 1 (Part 1) and [4] in Row 0 (Part 2) of your Place Value Tray. Combine the two parts to find the Whole.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): A number bond has a Whole of 8 and a known Part of 5 (5 + ? = 8). Find the missing Part 2.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 02 Lab: Part-Part-Whole & Number Bonds on the Place Value Tray

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: To make a Whole of 6, Part 1 is 2 (2 + ? = 6). What is the missing Part 2?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: In a number bond, Part 1 is 7 and Part 2 is 2. What is the Whole when you combine both parts?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Three parts join together: first combine 2 and 3, then add 4 to that sum. What is the Whole?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Ten birds sit on a fence (Whole = 10). Seven fly to a tree (Part 1 = 7). How many stay on the fence (Part 2)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

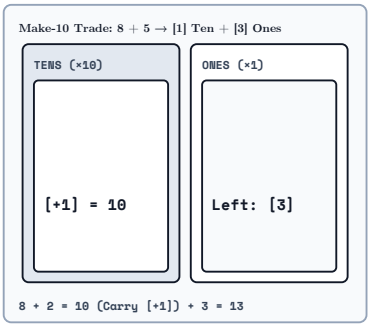
EXIT REFLECTION: How do adding Part 1 + Part 2 and subtracting Whole − Part 1 use the exact same number bond?

Complements to 10 & Make-10 Regrouping with Carry Tokens

Hands-On Model: Place Value Studio (Whole-Number Tray, Digit Blocks 0–9 & Gold [+1] Carry Tokens) • Standards: CCSS.MATH.1.OA.B.4, CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Complements to 10: Making Ten & Carry Trade):

Our base-10 system regroups every collection of 10 ones into 1 ten. Finding a number's complement to 10 (10 - 7 = 3) places the target base (10) on Shelf Y and the current ones digit (7) on Workbench X.



2. HANDS-ON BLOCKS & CONCEPT RULES: COMPLEMENTS TO 10 & GOLD [+1] CARRY RULES

- The Ones column holds single Digit Blocks [0] through [9]. The moment a column reaches 10, it regroups!
- 10-for-1 Carry Trade: When the Ones column reaches 10, place a Gold [+1] Carry Token into the Tens column and reset the Ones socket to [0].
- Complements to 10: To find how much more is needed to trigger a Gold [+1] carry, enter 10 [ENTER], then the digit, then [-].
- Bridging Through 10: Decompose the second addend to complete 10 first (e.g., 8 + 5 = 8 + 2 + 3 = 13).

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place the [7] Digit Block into the Ones socket of the Place Value Tray and hold a Gold [+1] Carry Token beside the Tens socket. Question: How many more ones must be added to [7] to reach 10 and trigger the Gold [+1] Carry Token? (7 + 3 = 10).

Verification Keystrokes: `1` `0` `ENTER` `7` `-`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`1`	1	Enter 1
`0`	10	Target ten (10) in X
`ENTER`	10	10 lifted to Y
`7`	7	Current Ones digit (7) in X
`-`	3	10 - 7 = 3: Adding 3 completes 10 and triggers the [+1] Carry!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): The Ones socket holds [6]. How many more ones are needed to complete a full 10 and trade for 1 Ten?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): To add 9 + 6 using the Make-Ten strategy, split 6 into 1 and 5 so you make 10 first (9 + 1 + 5). What is the sum?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 03 Lab: Complements to 10 & Make-10 Regrouping with Carry Tokens

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: You have 4 ones on your tray. How many more ones do you need to make a full 10 (4 + ? = 10)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Solve 7 + 5 by bridging through 10: first add 7 + 3 to make 10, then add the remaining 2.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: The Ones socket holds [9]. How many more ones are needed to complete 10 and regroup into 1 Ten?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Solve 8 + 6 by splitting 6 into 2 and 4 so you make 10 first (8 + 2 + 4).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

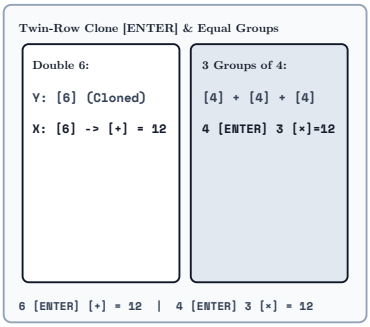
EXIT REFLECTION: Why does bridging through 10 (like 8 + 5 = 8 + 2 + 3) make adding across a ten easier?

Doubles, Near-Doubles & Equal Groups on the Place Value Tray

Hands-On Model: Place Value Studio (Two-Row Tray & Digit Blocks 0–9) • Standards: CCSS.MATH.1.OA.C.6, CCSS.MATH.2.OA.B.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Twin-Row Doubles: Adding Two Equal Groups):

Doubling a quantity means adding two identical copies of the same number (n + n = 2n). Seeing matching quantities in the top and bottom rows of the tray anchors mental doubles facts.



2. HANDS-ON BLOCKS & CONCEPT RULES: TWIN-ROW DOUBLES & NEAR-DOUBLES STRATEGIES

• Placing matching Digit Blocks in Row 1 (top) and Row 0 (bottom) of the Place Value Tray models a physical Double (n + n).

• Near-Doubles: When two addends differ by 1 (such as 7 + 8), double the smaller number first (7 + 7 = 14) and add 1 more (14 + 1 = 15).

• On the SC-6, pressing [ENTER] duplicates the number on Line X onto Line Y so pressing [+] checks any double in 3 keystrokes.

3. PARTNER / TEACHER STACK CHECK: VERIFYING OUR BLOCKS ON THE SC-6

Hands-On Setup: Place a [6] Digit Block in Row 1 (top) and a matching [6] Digit Block in Row 0 (bottom) of the Ones column on the Place Value Tray. Question: Observe the matching rows: 6 + 6 = 12 (1 Ten and 2 Ones on the tray).

Verification Keystrokes: [6] [ENTER] [+]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Place [7] in both rows of your Place Value Tray to build a twin-row double. What is the sum of two 7s (7 + 7)?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Use the known double 6 + 6 to solve the near-double 6 + 7 (double 6, then add 1).

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 04 Lab: Doubles, Near-Doubles & Equal Groups on the Place Value Tray

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with your Place Value blocks & draw in Box 1. (2) Write the two quantities & math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check with your partner or teacher on the SC-6!

Investigation 1: Two matching rows on the Place Value Tray each hold 8. What is double 8 (8 + 8)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Use the double 8 + 8 to solve the near-double 8 + 9 (double 8, then add 1).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: A carton has two equal rows of 9 eggs. What is double 9 (9 + 9)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Use the double 5 + 5 to solve the near-double 5 + 6 (double 5, then add 1).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+ or −):

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

EXIT REFLECTION: How can you use the twin-row double 7 + 7 = 14 to find 7 + 8 mentally in one step?

UNIT 05 GUIDED CONCEPT LAB • GRADES 3–4 (SC-6)

The 4-Register Stack: Four Operations & Automatic Lift

Hands-On Model: Place Value Blocks & 4-Register Stack • Standards: CCSS.MATH.3.OA.A.1, CCSS.MATH.3.OA.A.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Multi-Step Word Problems Without Parentheses):

In a multi-step expression like (3 × 4) + (2 × 5), each product is a self-contained quantity that must be evaluated before the two groups can be added. On the 4-level stack, the first product (12) waits safely on Shelf Y while the second product (10) is formed on X, requiring zero parentheses.



Place Value Studio (p. vi) • 10-for-1 Carry Trade

2. HANDS-ON BLOCKS & CONCEPT RULES: 4-REGISTER STACK ANATOMY

- Four registers: T (top), Z (third), Y (second/operand), X (screen/bottom).
- ENTER duplicates X into Y and pushes older values up into Z and T.
- Two-number operators (+, -, ×, ÷) combine Y and X, dropping Z to Y and T to Z.
- Automatic Lift: Entering a new number after an operation automatically pushes the result to Y!

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display. Question: Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

Verification Keystrokes: [8] [ENTER] [3] [-] [2] [×]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A baker bakes 15 cookies, sells 6 of them, and packs the remaining cookies equally into 3 bags. How many cookies go in each bag?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Evaluate (24 ÷ 6) + 15 by solving the division group first and then adding 15 to that result.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 05 Lab: The 4-Register Stack: Four Operations & Automatic Lift

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your step-by-step paper math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

Investigation 1: Solve (14 - 5) × 3 by finding the difference inside the parentheses first, then multiplying by 3.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Solve 30 ÷ (2 + 3) by adding the divisor group (2 + 3) first, then dividing 30 by that sum.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Evaluate (9 + 7) ÷ (8 - 4) by finding the numerator sum and denominator difference first, then dividing.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: A farmer collects 20 eggs, sets aside 2 cracked eggs, and divides the rest equally among 6 cartons. How many per carton?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

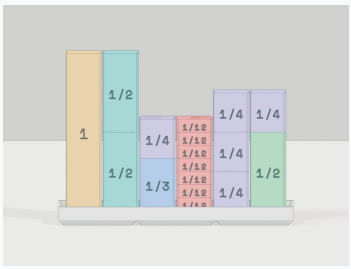
EXIT REFLECTION: In (15 − 6) ÷ 3, why don't we need parentheses when we compute the inner step (15 − 6) first?

Fraction Studio: Height Equivalence & Two-Dot Notation

Hands-On Model: 84 mm Fraction Towers & Twin Comparator Well • Standards: CCSS.MATH.4.NF.A.1, CCSS.MATH.4.NF.B.3

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Mixed Numbers & Height Equivalence: 3/4 + 2 1/2):

Fractions can only be added or subtracted when they share a common unit (denominator), and mixed numbers combine whole units with fractional parts. On the stack, two-dot notation ([.] n [.] d) treats each fraction or mixed number as a single unified value on Y and X.



Fraction Studio (p. vii) • 84 mm Flush Towers

2. HANDS-ON BLOCKS & CONCEPT RULES: TWO-DOT FRACTION ENTRY & TOWER EQUIVALENCE

- One Whole tower stands 84.0 mm tall, dividing cleanly into halves (42 mm), thirds (28 mm), fourths (21 mm), sixths (14 mm), eighths (10.5 mm), and twelfths (7 mm).
- Twin Comparator Well: Stack equivalent fraction towers side-by-side against the 84.0 mm Whole tower to verify equivalence by sight and touch.
- Two-dot entry format: [WHOLE] [.] [NUMERATOR] [.] [DENOMINATOR] (or [a ^{b}/c] on SC-6).
- Proper fractions have 0 whole: press [.] [3] [.] [4] for 3/4.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Setup: Place three 1/4 blocks (3 × 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm). Question: Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

Verification Keystrokes: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Stack a 1/2 tower (42 mm) and a 1/4 tower (21 mm) together. How many 1/4 blocks match their combined height, and what is 1/2 + 1/4?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Find 2/3 of 3/8 by multiplying (2/3) × (3/8) and simplifying to lowest terms. Which unit fraction tower matches the product?

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 06 Lab: Fraction Studio: Height Equivalence & Two-Dot Notation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your step-by-step paper math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

Investigation 1: Add 1/3 + 1/6 by trading 1/3 for two 1/6 blocks, then simplify the sum to lowest terms.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Subtract 1 1/2 - 3/4 by trading the whole and half into fourths (6/4 - 3/4). What fraction remains?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). Which is smaller, and what is 3/8 written as a decimal (3 ÷ 8)?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Find the area of a rectangle measuring 2/3 unit by 3/4 unit by multiplying (2/3) × (3/4) and simplifying.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

EXIT REFLECTION: Using the 84 mm Fraction Towers, explain why 1/2 + 1/4 equals 3/4 and NOT 2/6.

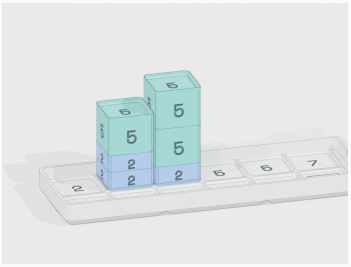
UNIT 07 GUIDED CONCEPT LAB • GRADES 5–6 (SC-6)

Postfix Evaluation, Factor Trees & Chained Operations

Hands-On Model: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack • Standards: CCSS.MATH.5.OA.A.1, CCSS.MATH.6.EE.A.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Chained Operations: Evaluating (a + b) × (c - d) from the Inside Out):

Written algebra uses parentheses to show which inner operations must happen first. Evaluating from the inside out follows true mathematical structure: compute the first branch (6 + 1 = 7), hold that result while computing the second branch (9 - 5 = 4), and multiply the two branch values together.



Prime Factor Studio (p. viii) • Log Prime Towers

2. HANDS-ON BLOCKS & CONCEPT RULES: INSIDE-OUT EVALUATION & PRIME FACTOR TREE RULES

- Order of operations works from the inside out: evaluate inner groups, parentheses, and factor branches before combining them.
- Every composite number decomposes into a unique tower of prime factors (e.g., 28 = 2 × 2 × 7).
- In nested fractions such as (a − b) ÷ (c + d), the fraction bar groups the numerator and denominator as separate sub-problems.
- When evaluating two separate branches like (a + b) × (c − d), solve each branch first and then multiply their results.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Setup: Use Prime Factor Studio blocks in the 12-socket comparator lid to model the two factor branches: (6 + 1) = [7] prime block and (9 - 5) = [2] + [2] = 4 tower, then combine their heights for 7 × 4 = 28. Question: Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined product: 7 × 4 = 28 (prime tower 2 × 2 × 7).

Verification Keystrokes: `6` `ENTER` `1` `+` `9` `ENTER` `5` `-` `×`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`6`	6	Value a (6) placed into Register X
`ENTER`	6	ENTER lifts 6 into Register Y
`1`	1	Value b (1) placed into Register X
`+`	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
`9`	9	Value c (9) entered into X; 7 lifted to Y
`ENTER`	9	ENTER lifts 9 into Z and 7 into T
`5`	5	Value d (5) placed into Register X
`-`	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y
`×`	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Evaluate (8 + 2) × (14 - 9) by finding the value of each parenthesized group first, then multiplying the two results.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Warm-Up B (Extension): Evaluate (15 - 5) ÷ (3 + 2) by computing the numerator and denominator separately, then dividing.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-6 (Key p. 25)**

Unit 07 Lab: Postfix Evaluation, Factor Trees & Chained Operations

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your step-by-step paper math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

Investigation 1: Evaluate (4 + 3) × (10 - 6) by solving each parenthesized group first.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 2: Evaluate (20 - 4) ÷ (2 × 4) by finding the numerator and denominator first.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 3: Evaluate (18 - 6) ÷ (1 + 2) and write the prime factorization of the numerator.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product before adding 5.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-6 Stack

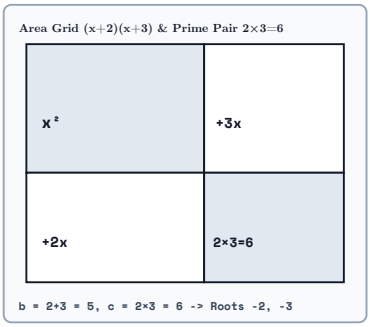
EXIT REFLECTION: In (8 + 2) × (9 − 4), where does the first branch result wait while you solve the second branch?

Algebraic Expressions, Base-x Polynomials & Quadratic Factoring

Hands-On Model: Prime Factor Studio (PRIME-01) & Algebraic Area Diagram • Standards: CCSS.MATH.7.EE.A.1, CCSS.MATH.8.EE.C.7

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks):

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$).



2. VISUAL MODEL & CONCEPT RULES: QUADRATIC AREA MODEL & PRIME FACTORING RULES

- A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0.
- To factor $x^2 + bx + c$ into $(x + p)(x + q)$, use Prime Factor blocks to factor c into two numbers p and q whose product is c and whose sum is b.
- When c is negative, p and q have opposite signs; when c is positive and b is negative, both p and q are negative.
- Zero-Product Property: The roots of $(x + p)(x + q) = 0$ are $x = -p$ and $x = -q$ (which you can check on the calculator with [POLY]).

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$. Question: Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots x = -2 and x = -3.

Verification Keystrokes: 1 ENTER 5 ENTER 6 g POLY

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
1	1	Coefficient a = 1 entered into X
ENTER	1	1 lifted to Y
5	5	Coefficient b = 5 entered into X
ENTER	5	b in Y, a in Z
6	6	Coefficient c = 6 entered into X
g POLY	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Factor $x^2 + 7x + 12$ by decomposing 12 with Prime Factor blocks to find the factor pair whose sum is 7, and find both roots of $x^2 + 7x + 12 = 0$.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Factor $x^2 - 5x + 6$ into two binomials by finding two negative factors of +6 whose sum is -5, and state both roots.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 08 Lab: Algebraic Expressions, Base-x Polynomials & Quadratic Factoring

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Factor $x^2 + 6x + 8$ by decomposing 8 into a factor pair that sums to 6, and find the roots.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Factor $x^2 - 4$ (difference of two squares, where the linear coefficient $b = 0$).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Factor $x^2 + 5x + 4$ into two binomials and state the solutions to $x^2 + 5x + 4 = 0$.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs and state both roots.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

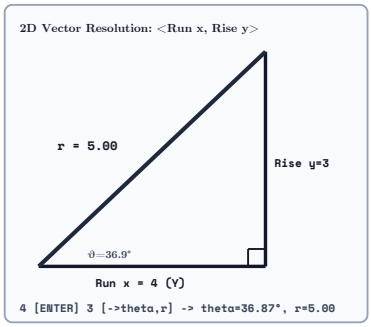
EXIT REFLECTION: For $x^2 + bx + c = (x + p)(x + q)$, how are the numbers p and q related to b and c?

Trigonometry, Vectors & Polar/Rectangular Coordinates

Hands-On Model: Right-Triangle & 2D Vector Diagram (Optional Ratio Check: Fraction Studio) • Standards: CCSS.MATH.HSG.SRT.C.8, CCSS.MATH.HSN.VM.A.1

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Vector Magnitude & Direction: Rectangular to Polar Coordinates):

Every 2D vector or right triangle relates two perpendicular legs (dx, dy) to a hypotenuse r and angle ϑ via $r = \sqrt{(x^2 + y^2)}$ and $\tan(\vartheta) = y/x$. Keeping horizontal component dx in Register Y and vertical component dy in Register X pairs the two coordinates naturally.



2. VISUAL MODEL & CONCEPT RULES: RIGHT-TRIANGLE RATIOS & 2D VECTOR DECOMPOSITION RULES

- Trigonometric functions are ratios of right-triangle sides: sin(ϑ) = Opposite / Hypotenuse, cos(ϑ) = Adjacent / Hypotenuse, tan(ϑ) = Opposite / Adjacent.
- Every 2D vector v = <dx, dy> forms a right triangle with magnitude $r = \sqrt{(dx^2 + dy^2)}$ and direction angle ϑ = arctan(dy / dx).
- To add two vectors A = <x1, y1> and B = <x2, y2>, add their horizontal components (x1 + x2) and vertical components (y1 + y2) separately.
- Stack Check: Place horizontal component dx in Y and vertical component dy in X, then press [f:] [→ϑ,r] to verify ϑ and r!

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Draw a right triangle with horizontal run dx = 4 (Register Y) and vertical rise dy = 3 (Register X) to model the 3-4-5 vector. Question: Run = 4, Rise = 3. Predict magnitude: $r = \text{sqrt}(4^2 + 3^2) = 5$. Predict angle: arctan(3/4) ~ 36.87°.

Verification Keystrokes: [4] [ENTER] [3] [↶] [▸ϑ,r] [x⇄y]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸ϑ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A surveyor walks 60 m East (dx = 60) and 80 m North (dy = 80). Use the Pythagorean theorem and tangent ratio to find the straight-line distance r and angle ϑ north of east.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector <x_net, y_net> and its magnitude r.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 09 Lab: Trigonometry, Vectors & Polar/Rectangular Coordinates

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Find the magnitude r and direction angle ϑ for the vector <12, 5>.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find the hypotenuse of a right triangle with legs of length 7 and 24.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: In a 30°-60°-90° triangle, the side opposite 30° is half the hypotenuse (42 mm vs. 84 mm). What is sin(30°)?

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: A vector has magnitude r = 10 at angle ϑ = 45°. Find its horizontal (x = r cos ϑ) and vertical (y = r sin ϑ) components.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How are the horizontal run (dx) and vertical rise (dy) of a vector related to its magnitude r?

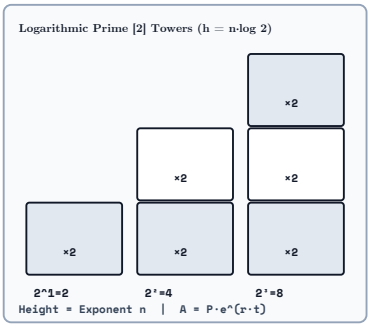
UNIT 10 GUIDED CONCEPT LAB • GRADES 10–12 (SC-32)

Exponential Growth, Compound Interest & Logarithms

Hands-On Model: Prime Factor Studio (PRIME-01 Logarithmic Blocks) & Growth Curve • Standards: CCSS.MATH.HSF.LE.A.1, CCSS.MATH.HSF.LE.A.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Continuous Compounding & Euler's Constant e):

Exponential growth P·e^(rt) compounds continuously by scaling a base quantity according to an inner rate-time product (r × t). Evaluating from the inside out—computing the exponent r × t first, raising the base, and multiplying by P—follows the exact mathematical hierarchy without parentheses.



2. VISUAL MODEL & CONCEPT RULES: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Discrete exponential growth multiplies by a constant factor each period: A = P × b^t (or A = P × (1 + r/n)^(n·t)).
- Continuous exponential growth follows A = P × e^(r·t), where Euler's constant e ~ 2.71828 is the base of natural growth.
- Logarithms invert exponentials (ln(e^x) = x) and turn multiplication into addition: ln(a × b) = ln(a) + ln(b).
- To find continuous doubling time, solve e^(r·t) = 2 by taking the natural logarithm: t = ln(2) / r ~ 0.6931 / r.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Stack Prime Factor [2] blocks (15 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$). Question: Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).

Verification Keystrokes: `0` `.` `0` `7` `ENTER` `1` `0` `×` `eˣ` `1` `0` `0` `0` `×`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`0` `.` `0` `7`	0.07	Rate r = 0.07 entered in X
`ENTER`	0.07	Rate lifted to Y
`1` `0`	10	Time t = 10 entered in X
`×`	0.7	Exponent r × t = 0.70
`eˣ`	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
`1` `0` `0` `0`	1000	Principal P = 1000 (growth factor lifts to Y)
`×`	2013.75	Total accumulated amount A = $2,013.75!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A bacteria culture starts with P = 500 cells and triples every hour (A = 500 × 3^t). Calculate the population after t = 4 hours.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): A radioactive isotope decays according to N(t) = N_0 · e^(-0.035t). Find its half-life t = ln(2) / 0.035 to two decimal places.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 10 Lab: Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate e^2 to 4 decimal places.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find the continuous doubling time t = ln(2) / r for an investment growing at r = 8% (0.08).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate 10^3.5 (note that 10^3 < 10^3.5 < 10^4).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Solve 10^x = 500 for x using the base-10 logarithm log10(500).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

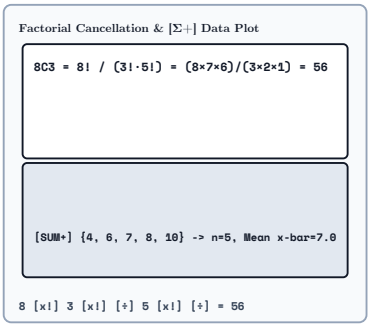
EXIT REFLECTION: Why is the natural logarithm [LN] the inverse operation needed to solve for doubling time t?

Combinatorics, Probability Ratios & Series Accumulation

Hands-On Model: Prime Factor Studio (PRIME-01 Factorial Cancellation) & Statistical Data Plot • Standards: CCSS.MATH.HSS.CP.B.9, CCSS.MATH.HSA.SSE.B.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Combinations vs. Permutations: Subsets & Prime Cancellation):

Combinations nCr = n! / (r!(n-r)!) count unordered selections by dividing ordered permutations by the r! duplicate arrangements of the chosen group. Canceling 5! from 8! leaves (8 × 7 × 6) / 3! = 56.



2. VISUAL MODEL & CONCEPT RULES: COMBINATIONS, PERMUTATIONS & SAMPLE STATISTICS RULES

- Factorial n! = n × (n − 1) × ... × 1 counts the ways to arrange n distinct items; Prime Factor blocks show how common factors cancel in n! / r!.
- Permutations (order matters): nPr = n! / (n − r)! counts ordered arrangements of r items chosen from n.
- Combinations (order does not matter): nCr = n! / (r! × (n − r)!) divides out the r! duplicate orderings.
- Sample Mean: x-bar = (sum of values) / n. You can check factorials with [f:][x!] and sample statistics with [Σ+].

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / (3 × 2 × 1) = 8 × 7 = 56. Question: How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 56.

Verification Keystrokes: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): A lottery draws 4 winning numbers from a pool of 10 (order does not matter). Simplify 10C4 = 10! / (4! × 6!) by canceling 6! first, then check your result.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Given the four data points {12, 15, 18, 21}, find the sample size n, the total sum of the values, and the sample mean x-bar.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 11 Lab: Combinatorics, Probability Ratios & Series Accumulation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate 6! (the number of ways to arrange 6 distinct books on a shelf).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: How many ways can 5 runners finish in 1st, 2nd, and 3rd place? (Compute 5P3 = 5! / 2!).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: How many 2-person partnerships can be formed from 7 students? (Compute 7C2 = 7! / (2! × 5!)).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Find the mean x-bar of the two data values 5 and 15.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

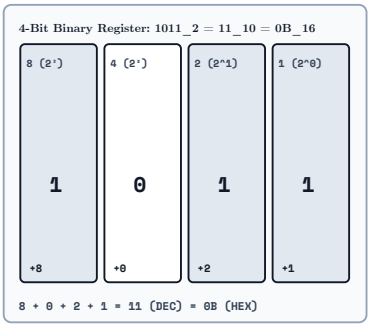
EXIT REFLECTION: Why does a Combination C(n, r) produce fewer outcomes than a Permutation P(n, r)?

Digital Logic, Base Conversions & Computer Architecture

Hands-On Model: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns) • Standards: CCSS.MATH.PRACTICE.MP5, CCSS.MATH.HSA.SSE.A.1

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Positional Number Systems: Decimal, Binary & Hexadecimal):

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) or 16 instead of powers of 10. Every 4-bit binary group (0000 to 1111) maps to exactly one hexadecimal digit (0 to F).



2. VISUAL MODEL & CONCEPT RULES: POSITIONAL NUMBER BASES & BITWISE LOGIC RULES

• Binary (Base 2) uses powers of 2 (..., 8, 4, 2, 1) with digits 0 and 1; Hexadecimal (Base 16) groups 4 binary bits into a single digit 0–9 or A–F.

• Bitwise AND (&): output bit is 1 only when BOTH input bits in that column are 1 (used for masking and network subnets).

• Bitwise OR (|): output bit is 1 when EITHER input bit is 1. Bitwise XOR (^): output bit is 1 when the two input bits DIFFER.

• Stack Check: Press [f:] [BASE] to verify conversions between DEC, HEX, BIN, and OCT, or use [AND], [OR], [XOR] on two integers.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits. Question: Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary 1011_2 = 8 + 0 + 2 + 1 = 11_10. In hexadecimal (A=10, B=11), 11_10 = 0B_16.

Verification Keystrokes: [1] [1] [↖] [BASE]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[1] [1]	11	Enter decimal integer 11 in DEC mode
[↖] [BASE]	HEX 0B	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↖] [BASE]	BIN 1011	Press [f:] [BASE]: screen displays Binary 1011!
[↖] [BASE]	OCT 13	Press [f:] [BASE]: screen displays Octal 13!
[↖] [BASE]	DEC 11	Press [f:] [BASE]: returns to Decimal 11

WARM-UP CHECK: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Warm-Up A (Core): Express decimal 255 as a sum of powers of 2 (128 + 64 + 32 + 16 + 8 + 4 + 2 + 1) to write its 8-bit binary and 2-digit hexadecimal representations.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Warm-Up B (Extension): Compare 12 (1100_2) and 3 (0011_2) column by column to find the result of 12 OR 3 in both binary and decimal.

1st Quantity / Step: _____ **2nd Quantity / Step:** _____ **My Paper Answer:** _____ **[ ] Checked on SC-32 (Key p. 25)**

Unit 12 Lab: Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the geometric/coordinate model in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Decompose decimal 42 into powers of 2 (32, 16, 8, 4, 2, 1) to write its 6-bit binary representation.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Convert hexadecimal 2A_16 to decimal using place value (2 × 16^1 + 10 × 16^0).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate the bitwise AND of 15 (1111_2) and 7 (0111_2) in binary and decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate the bitwise XOR of 12 (1100_2) and 10 (1010_2) in binary and decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: Why does a single hexadecimal digit (0–F) correspond to exactly 4 binary bits (one nybble)?

Student Self-Check Answer Key

Check your Warm-Up and Investigation answers after completing each unit. Full step-by-step traces are in Book 1.

Unit 01: Cardinality, Counting & Step Accumulation (Pre-K – Kindergarten)
Warm-Up A: 5 ([5] Digit Block; 4 + 1 = 5) | B: Row 1 has 4 more (6 - 2 = 4)

Investigation 1: 5 • **Investigation 2: 3**
Investigation 3: 0 (Zero) • **Investigation 4: 4**

Unit 02: Part-Part-Whole Number Bonds & Missing Addends (Grades K–1)
Warm-Up A: 9 ([9] Digit Block; 5 + 4 = 9) | B: Missing Part is 3 (8 - 5 = 3)

Investigation 1: 4 • **Investigation 2: 9**
Investigation 3: 9 • **Investigation 4: 3**

Unit 03: Complements to 10 & Gold [+1] Carry Regrouping (Grade 1)
Warm-Up A: 4 ones (10 - 6 = 4) | B: 15 (9 + 1 = 10, then 10 + 5 = 15)

Investigation 1: 6 • **Investigation 2: 12**
Investigation 3: 1 • **Investigation 4: 14**

Unit 04: Twin-Row Place Value Tray: Doubles & Equal Groups (Grades 1–2)
Warm-Up A: 14 ([1] Ten and [4] Ones on the tray; 7 + 7 = 14) | B: 13 (6 + 6 = 12, then 12 + 1 = 13)

Investigation 1: 16 • **Investigation 2: 17**
Investigation 3: 18 • **Investigation 4: 11**

Unit 05: Place Value & The 4-Level Stack Elevator (Grades 3–4)
Warm-Up A: 3 cookies per bag ((15 - 6) ÷ 3 = 3) | B: 19 (24 ÷ 6 = 4, then 4 + 15 = 19)

Investigation 1: 27 • **Investigation 2: 6**
Investigation 3: 4 • **Investigation 4: 3 eggs**

Unit 06: Fraction Studio: 84 mm Towers & Two-Dot Entry (Grades 4–5)
Warm-Up A: 3/4 (three 21 mm fourth-blocks = 63 mm) | B: 1/4 (6/24 = 1/4, height 21 mm)

Investigation 1: 1/2 (42 mm) • **Investigation 2: 3/4 (63 mm)**
Investigation 3: 0.375 • **Investigation 4: 1/2 (42 mm)**

Unit 07: Postfix Trees, Order of Operations & Prime Factors (Grades 5–6)
Warm-Up A: (8 + 2) = 10 and (14 - 9) = 5; 10 × 5 = 50 (Prime factors: 2 × 5 × 5) | B: Numerator = 10, Denominator = 5; 10 ÷ 5 = 2

Investigation 1: 7 × 4 = 28 • **Investigation 2: 16 ÷ 8 = 2**
Investigation 3: 12 ÷ 3 = 4 (12 = 2×2×3) • **Investigation 4: 5 + 12 = 17**

Unit 08: Base-x Polynomials, Prime Factor Pairs & Roots (Grades 7–8)
Warm-Up A: 3 × 4 = 12 and 3 + 4 = 7; factored form (x + 3)(x + 4), roots x = -3 and x = -4 | B: (-2) × (-3) = +6 and (-2) + (-3) = -5; factored form (x - 2)(x - 3), roots

Investigation 1: (x + 2)(x + 4); roots -2, -4 • **Investigation 2: (x - 2)(x + 2); roots 2, -2**
Investigation 3: (x + 1)(x + 4); roots -1, -4 • **Investigation 4: (x - 2)(x - 3); roots 2, 3**

Unit 09: Coordinate Geometry, Trigonometry & Vectors (Grades 9–11)
Warm-Up A: r = √(60² + 80²) = 100 m; ϑ = arctan(80/60) ˜ 53 | B: Resultant <3, 6>; Magnitude r = √(3² + 6²) = √45 ˜ 6

Investigation 1: r = 13, ϑ = 22.62° • **Investigation 2: r = √(49 + 576) = 25**
Investigation 3: 42 / 84 = 0.5 • **Investigation 4: x ˜ 7.07, y ˜ 7.07**

Unit 10: Exponents, Logarithms & Continuous Growth (Grades 10–12)
Warm-Up A: 3^4 = 81; 500 × 81 = 40,500 bacteria | B: t = ln(2) / 0

Investigation 1: 7.3891 • **Investigation 2: 8.66 years**
Investigation 3: 3162.28 • **Investigation 4: 2.6990**

Unit 11: Combinatorics, Probability Ratios & Series Statistics (Grades 11–12 / AP)
Warm-Up A: (10 × 9 × 8 × 7) / (4 × 3 × 2 × 1) = 5040 / 24 = 210 combinations | B: Count n = 4, Sum = 66, Mean x-bar = 66 / 4 = 16

Investigation 1: 6 × 5 × 4 × 3 × 2 × 1 = 720 • **Investigation 2: 5 × 4 × 3 = 60**
Investigation 3: (7 × 6) / 2 = 21 • **Investigation 4: (5 + 15) / 2 = 10**

Unit 12: Digital Logic, Base-N & Computer Architecture (Grades 9–12 / AP CS)
Warm-Up A: Binary: 11111111_2 (eight 1s); Hexadecimal: FF_16 (15 × 16 + 15 = 255) | B: 1100_2 OR 0011_2 = 1111_2 = 15_10

Investigation 1: 32 + 8 + 2 → 101010_2 • **Investigation 2: 32 + 10 = 42**
Investigation 3: 0111_2 = 7 • **Investigation 4: 0110_2 = 6**

Certificate of Hands-On Stack Mastery

Awarded for completing Units 01–12 of the StackCalc Hands-On Math Lab Workbook.

CERTIFICATE OF HANDS-ON MATHEMATICS MASTERY

This certifies that: _____

has completed the 12 Hands-On Math Lab investigations from Counting & Number Bonds through Fractions, Algebraic Factoring, Trigonometry, Logarithms, Probability, and Digital Logic—building physical block models first and verifying every solution on the 4-level StackCalc RPN stack (T, Z, Y, X).

[ ] Elementary Stack Explorer (Units 01–07 • SC-6) **[ ] Scientific Stack Master (Units 08–12 • SC-32)**

Teacher / Parent Signature: _____ Date: _____

ESSENTIAL SC-6 & SC-32 KEYCAP SEQUENCES TO REMEMBER

Doubling Any Number in Register X (e.g., 2 × 7 = 14)	
Parenthesis-Free Chained Math: (8 + 2) × (9 − 4) = 50	
Two-Dot Fraction Addition: 3/4 + 1/2 = 1 1/4	
Quadratic Roots of $x^2 - 5x + 6 = 0$ (Roots: 2, 3)	
Pythagorean Hypotenuse of 3 and 4 → 5	